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Modeling, evaluation and metrics performance of the SyncLMKD in distributed kinematics variations

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Modeling the Digital Twin (DT) is an important resource for accurately representing the physical entity, enabling it to deliver functional services, meet application requirements, and address the disturbances between the physical and digital realms. This article introduces the Log Mean Kinematics Difference Synchronization (SyncLMKD) to measure the kinematic variations distributed among Digital Twin elements to ensure symmetric values relative to a reference. The proposed method employs abductive reasoning and draws inspiration from the Log Mean Temperature Difference (LMTD). The SyncLMKD is applied to measure kinematic variations among mobile robot entities in a DT representation, and it operates following the progression of displacement. A suitable synchronization technique was also developed for experiments based on this method. The main advantage of the SyncLMKD is its high precision in displacement measurement and predictability of distances between counterparts and the dynamic target of the Digital Twin, all while requiring low computational effort. The approach was tested with robots positioned at various locations and speeds to achieve synchronization among them. The SyncLMKD method demonstrated a precision of 10^{-4} in measuring the distances between elements, achieving synchronized movement of the counterparts with speed adjustments facilitated by the synchronization technique, with percentages ranging from 150 to 200%.

Keywords Modeling, Digital twin, SyncLMKD method, Kinematic variations, Synchronization technique

The concept of the Digital Twin (DT) has evolved significantly over the years, initially emerging from the fields of aerospace and manufacturing. Digital Twins serve as virtual representations of physical entities, enabling real-time monitoring, simulation, and optimization of systems. Integrating physical and digital worlds has found applications across various industries, including automotive, healthcare, and smart cities, significantly enhancing decision-making processes and operational efficiencies.

A major challenge associated with Digital Twins is bridging the gap between simulation and reality; simulations only sometimes reliably reflect their real-world counterparts. Specifically, achieving consistent adherence to predefined paths toward target positions proves difficult when representing mobile robots and their interactions with both physical and virtual environments.

In the synchronization of Digital Twins, there is no competition between workspaces, nor is there a need for precise tuning of choreographed movements. The primary goal is to create an accurate digital representation of the real world. Thus, the capability of the Digital Twin to function as a decoupled entity within service compositions-whether choreographed or orchestrated-is essential for effective commissioning. The service composition must operate independently within the virtual environment, and similarly, the physical service composition should function without dependence on synchronized interactions with the Digital Twin.

Consequently, adjustments to the kinematic movements for synchronization between mobile robot entities within the Digital Twin should be made dynamically when disturbances are detected rather than adhering to a rigid "recipe" approach typically used for choreographed movements. This flexibility is crucial for maintaining effective synchronization, allowing both the virtual and physical components to respond seamlessly to real-time conditions.

The mobile robot cannot follow the planned path, hence on account of environmental factors influence or to the errors of coordination of several parts of the robot motion get compromised[1–3]. At the same time, the lack

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of data feedback of most robots during the movement makes it impossible to determine where errors take place and correct the robot's workspace making it impossible to work in locals with high operational precision[4,5]. The nonlinear control techniques take advantage of the knowledge or system parameters to develop the behavior of the mobile robot[6]. Furthermore the control system is responsible for coordinating data entry and planning the movement of the robot in order to make it move properly[7]. In recent years, researchers have worked on dynamic path planning methods based on ideal control. For that they take in consideration dynamic and the physical characteristics of the robot to plan the trajectory[8]. The fractional order controls can be utilized to increase the precision of the robotic system to such a more realistic degree than the ones given by the PID traditional controller (proportional-integral-derivative controller)[9]. The problem of the trajectory tracking is defined by seeking such a control law that the difference between the planned trajectory and the real trajectory followed by the mobile robot be zero[10]. The use of nonlinear models which integrate lateral and longitudinal dynamics along with the orientation seeks to determine the control entries needed to follow or to track the desired trajectory which was previously planned for the mobile robot avoiding at the same time disturbances due to unmodeled dynamic effects as friction and noise[11].

Factories 4.0, Industries 4.0 and mainly logistics distribution centers have mobile robots and for the tendency to digitize their processes onto the Digital Twin will need to apply different levels of synchronization considering goods, parts, equipments and others pass through conveyor belts with fixed sensing controllers by static robotic manipulators and even by the free displacement of cargo by the environment. All those automated processes need to be digitized onto virtual models with high or low degree of synchronization as that representativity, real environment into the virtual one, is known to be almost impossible to be recreated in its whole aspect. In that aspect, the Digital Twin employs mathematical solutions, physical, data warehouse, blockchain and ones which mix several others, as hybrid ones, through softwares with the intention to compensate for discrepancies and disturbances between the virtual models and their physical counterparts. By that it is not enough that the models be only reconfigured for the commissioning stage, full or partial, they also need to be sensible for discrepancies and disturbances which the physical environment can have. Numerous trails are undertaken previously by using AI techniques and sets of data process and in turn are situations that can cause problems to the automated process although those techniques can be valid concerning to sensibility and adaptation to the process, they depend on training and it generates extra time and a lot of processing, that is, when one think of the synchronization property this is counterproductive because the real environments are dynamic and the synchronism must take place instantly not a posteriori. Other problem on those training trials for the synchronization with the Digital Twin is the usage most of time of raw data, rigged data, noises, dirty data, ideally it would be important to filter them but that task is not practical and nor safe considering the view of there is not dataset crossing and the detection of anomalies cannot be detect and therefore the chance of those training trials go wrong is big[11]. Once shown that, techniques which apply direct analytical methods and that solve problems numerically are more interesting from the point of view of filtering data, for the calculation right at the entry, also by making all the adjustment of discrepancies and disturbances online.

This work is the result of other three ones undertaken previously. Paper[12] analyzed a set of data to measure the heat transfer in data processing units and the same data were used at the present modeling of the SyncLMKD method. Paper[13] refers to simulation architecture and the first outcomes about the algorithmic control of synchronization between the counterparts of the Digital Twin. Paper[14] focused on the development of the technique of synchronization based on the SyncLMKD method. The work detailed the methodology employed to reach the synchronization of the counterparts while reaching the target by using dynamic displacement. It is highlighted that the numerical modeling of the SyncLMKD method, its evaluation for distributed kinematic variations measurement and performance metrics were not mentioned previously and therefore and the percentage ratio of the SyncLMKD method with the predictability of the distances will be presented on the present paper. Outlining this article, in "Related works" section, researches related to synchronization of mobile robots for DT are presented. "The LMTD method" section shows the renowned method of heat transfer. The "LMTD modeled" section shows the data processing and verisimilitudes found among problems and solutions which can be generalized for the Digital Twin domain. The "LMTD remodeled" section shows an analysis of the set of data for inference of hypothesis according to abductive reasoning. "The SyncLMKD method" section presents the method inspired in remodeling and in control flow. "Evaluation" section poses the architecture of the simulation environment, the comparative analysis of the percentages and the predictability between distances. Moreover, "Analysis and discussion" Section highlights the limitations found during the research. In the "Performance metrics" section there is the layout presentation for synchronization efficiency metrics. Finally in the "Conclusion" section conclusions of the current work are provided.

Related works

Recent advancements have propelled the usage of Digital Twins, particularly in robotics. As the demand for greater precision and reliability in robot synchronization grows, researchers have increasingly explored methodologies to bridge the gap between physical robots and their digital counterparts. In this context, various studies have contributed to the development and refinement of Digital Twin technologies.

Kuts, Cherezova, Sarkans, and Otto[15] studied the precision of the digital environment. Their approach involved transferring the mathematical model to the digital environment by creating a precise virtual and scaled model of an industrial robot, which enhanced the level of the Digital Twin (DT) developed environment. However, despite using a parametric virtual model with the same scales as the real model, tests revealed a synchronization lag of 4 millimeters. It is noteworthy that the Jacobian matrix was not employed to adjust both linear and angular speeds; instead, the adjustments were made using the Unit3D engine frames themselves.

Liang, McGee, Menassa, and Kamat[16] developed the Pose Checking Algorithm (PCA) to verify the synchronization of two robotic arms. This algorithm utilizes the current virtual robot's pose, the current real

robot's pose, and the next robot's pose as input parameters. PCA calculates the difference between the virtual and real arms. If the difference exceeds a predefined limit, the next joint angle is assigned to the current joint angles to ensure that the physical robot can reach the desired positions. Consequently, the trajectory must be planned to reflect the updated joint angles.

Müller, Jazdi, and Weyrich[17] identified three key steps to be implemented in both the cyber and physical domains: data acquisition, preprocessing, and transfer. The cyber domain produces synthetic data without noise, while the real domain generates noisy data ("dirt"). Due to the friction affecting the real mobile robot during its displacement, they observed that although a speed of 20 cm/s was sent by the controller to both robots, the physical robot achieved only 16 cm/s, as opposed to the 20 cm/s of the digital one. To address this nonlinearity, they utilized reinforcement learning, which does not assume a linear relationship between the required and actual speeds, allowing for "jumps" whenever position updates indicate that a deviation limit of 2 cm has been breached.

Liu et al.[18] employed genetic algorithms to map the trajectories of robots, which may optimize the navigation of mobile robots by reducing errors in moving the physical counterparts through the interaction of virtual and real data. This approach offers a method for facilitating the physical robot's learning within the digital domain.

The articles mentioned in this section present strategies to enhance synchronization in Digital Twin applications. One approach involves transferring the mathematical model to the digital environment, aiming for an accurate reflection of the real model. However, this method proves to be unfeasible, as the physical world is dynamic and complex, making it impossible to fully digitize. The Pose Checking Algorithm (PCA) is another method that only accounts for the difference between the next pose or angle for adjustments. This approach means that task planning, conducted sequentially until the target is reached, can lead to instability in control or result in undesirable "robotic" movements that are characterized by frequent stops and starts. Reinforcement learning, utilized to validate the model through repetitive simulations until improvements are observed, requires significant time and additional processing. Even in a controlled and stable environment, the use of genetic algorithms necessitates numerous iterations to achieve a gradual reduction in twinning errors in Digital Twin applications.

In contrast, this article introduces the SyncLMKD method to overcome these limitations by considering the global displacement of counterparts in real-time, in alignment with dynamic path planning. This method generates smooth and uniform adjustments to kinematic movements, facilitating flexible control through the decoupling of onboard controls in both counterparts as well as during path planning. Additionally, the synchronization technique offers a degree of predictability, enabling the anticipation of behavioral approximations between the Digital Twin entities. This predictive capability significantly enhances the overall effectiveness of synchronization in Digital Twin applications, ensuring a more reliable integration of physical and digital systems.

The LMTD method

Log Mean Temperature Difference (LMTD) was developed by Rudolf Clausius in the middle of the XIX century, as an extension of his second law of thermodynamic formulation. Originally it was developed for the efficiency of thermal machines calculation, but later it was applied to the heat transfer in continuous flow heat exchangers. The method is based on observing the heat transference range between two fluids which are in contact through a solid wall, and it is proportional to the temperature difference average of fluids. LMTD is a weighted average of log temperature differences of temperature between fluids along the length of heat exchangers. Although it was developed more than a century ago, it is widely used due to its precision and ease of use also on its applicability to a variety of continuous flow of heat exchangers, turning it into a valuable tool for engineers who work with heat transfer and exchangers projects. According to INCROPERA book[19] reference for heat and mass transfer base, the main two methods of heat exchangers analysis are: LMTD (Log Mean Temperature Difference) and ε-NUT (number of transfer units and effectiveness - ε). In any situation, both methods can be used and can supply equivalent outcomes.

Log Mean Temperature Difference (LMTD) is calculated based on temperatures of entering and exit of the heat exchangers according to the equation presented below.

$$LMTD = \frac{(T1c - T2h) - (T2c - T1h)}{Ln\frac{(T1c-T2h)}{(T2c-T1h)}} \tag{1}$$

where, $T1c$ ts the Temperature of the cold fluid entering;$T1h$ is the Temperature of the hot fluid entering; $T2c$ is the Temperature of cold fluid exit; and $T2h$ is the Temperature of hot fluid exit.

Calculated values of energy and LMTD, one can calculate the surface of thermal exchange according to equation

$$Q = U.A.LMTD, \tag{2}$$

where, U is the Heat transfer coefficient ($\mathrm{kW/m^2\,^\circ C}$), specific to each kind of equipment and given by the manufacturer; and A is the Thermal exchange area ($\mathrm{m^2}$).

The LMTD method is suitable when the target temperature is known. Although, it will not be possible simply to get an average from temperature difference because that average between hot and cold fluids varies along the

heat exchanger. So, using the og mean kinematics difference is the most suitable to calculate the heat transfer range. By affirming that, the heat transfer range is proportional to the difference of temperatures of fluids.

The LMTD Method is robust for the heat exchange calculation still for the following reasons:

1. Efficiency. Thermal efficiency is a capacity measure of the heat exchanger to transfer heat from one fluid to another and it can be used to evaluate the efficiency of that heat exchanger.
2. Accuracy: It is a precise analytical method to calculate the heat transfer. It takes into consideration the fluid temperature variation along its length which makes it more precise than the simplified methods that assume the fluid temperature to be constant.
3. Facility: The LMTD Method is relatively simple to use and it does not require software use or special devices. Calculations can be done by using a simple calculator.
4. Applicability: It can be used to calculate the heat transfer in such a variety of heat exchangers including plate heat exchangers, tube heat exchangers and shell and pipe heat exchangers simultaneously.
5. Validity: It is based on thermodynamic principles well established and it has been experimentally validated in a great number of heat exchangers application.For all those reasons and others not listed[20] the LMTD Method is widely used for the calculation of heat exchangers in continuous flow heat exchangers.

LMTD modeled

On his classical book "Productive Thinking", Wertheimer[21] linked the understanding to the transference by affirming that understanding is important because it gives the capacity to generalize a problem solution to apply it to other problem, that way, there are problems which modeled LMTD solve about heat transfer and whose verisimilitude of problems and solutions can be applied to solve problems on kinematics for mobile robots in dynamic environments, mainly when it treats on mirroring for the application of Digital Twin.

Some premises are raised to support the argument made.

- Heat transfer occurs dynamically in "N" points of interface of heat exchanger and for that logarithmic propagation is necessary to observe the tendency to transference similarly the displacement of the physical and virtual robot trajectory finds "N" nonlinear factors along the planned trajectory which via the logarithmic can demonstrate the dynamic of the position transference;
- The thermal transfer between the heat source and the cold fluid occurs between the interface which separates them similarly the application of the Digital Twin also has a data exchange area "interface" between the simulation (virtual) and the environment of the real robot (physical);
- The modeled LMTD for heat transference in semiconductors, which will be presented further, nas a heat source as known reference for its calculation likewise on SyncLMKD proposition finds a known reference at the planning path as entering for the calculation with the physical counterpart and the virtual one;
- The value resulting from the calculation of LMTD shows the efficiency of the heat exchanger to transfer the heat between fluids or then lower the heated surface temperature once considering SyncLMKD is possible to measure directly the efficiency of the controller in issue being them distinctive, as much at the physical robot as at the virtual one, being equals, the same controller for both counterparts;
- Logarithms can indicate tendencies along the period checked so the cooling system can foresee the necessity of the increasing of the cold fluid flow for cooling and keep the control over the temperature similarly the control flow proposed further allows to foresee the necessity of speed adjustment for the counterparts trajectory, basing on the logarithmic tendency also allows to configure a control margin which maintains the counterparts rigidly controlled or then with a fair relaxing through the synchronization.In addition to those detailed explanations above it is also necessary to evaluate the set of limited data to infer the hypothesis according to the abductive reasoning. Abductive reasoning is a scientific method used to infer the hypothesis onto the experiment, that is to say, plausible explanations from the limited data. It seeks to analyze the set of data to detect robust empirical irregularities or phenomena. Once detected those phenomena are explained by abductive inference of cause subjacent mechanisms. After positive judgments of initial plausibility of those explanatory theories, attempts are undertaken to elaborate the nature of the cause mechanisms at issue. That is done by building models of those mechanisms by analogy with relevant ideas in domains which are well comprehended[22].

To sum up abductive reasoning consists on three steps:

1. Seek elements which, although different, have the same characteristics when treated at the same method;
2. Evaluate the set of limited data to infer the hypothesis;
3. Make experiments in order to confirm the hypothesis.To corroborate that issue, that is the experimental analysis of the set of data, a set of data of a LMTD method modeled were analyzed, undertaken in research[12] which measured the heat transference in data processing units. An experimental apparatus was built to obtain the precise values of the fluids temperature as well as the values of heat source temperature.

The set of data analyzed on that modeling of LMTD method to measure the heat transference in processing units of data as *Tin* is the Temperature of cooling inlet fluid in the waterblock; *Tout* is the Temperature of cooling outlet fluid in the waterblock; *TCPU* is the Central processing unit temperature; and *Model* is the Heat tax transfer.

The difference of the mean log temperature Model is written as

$$Model = \frac{[(TCPU - Tin) - (TCPU - Tout)]}{Ln\left[\frac{(TCPU-Tin)}{(TCPU-Tout)}\right]}. \tag{3}$$

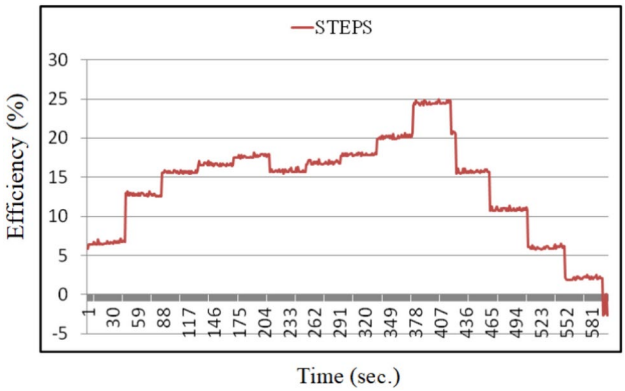


Fig. 1. LMTD modeled.

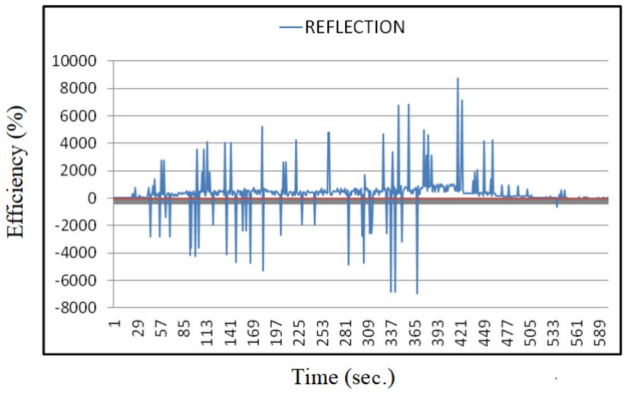


Fig. 2. LMTD remodeled.

As observed the analyzed set of data were the ones which differently were used for the calculation of equation 3 of Model for transferring heat in processing units.

Time was obtained second by second as well as respectively the inlet fluid temperature, the outlet fluid temperature and the heat source temperature, that is to say, the processor. In Fig. 1 the outcome of the Model is presented related to elapsed time, the steps observed occur due to internal drive cycles of the CPU cores. It was observed that after the test of stress of the processing unit be concluded occurs a systematic shutdown of the cores leading to shortening of the time of the cycle descend which was of 436 s to 581 s.

Analyzing the chat in Fig. 1 is possible to understand that the highest temperature of the processing unit was 53°C when the percentage of efficiency of the heat exchanger (copper block with microproducts from where the fluid drains) it reached about 25% of efficiency in a time of 419 s. Also when the processing unit temperature got 25°C, inferior to warm and cold fluids, the percentage inverted for -2% in a time of 594 s. Equation (3) the way it is allows only two subjacent inferences:

1. When the value of the heat source is quite above from the warm and cold fluids, the percentage will be high;
2. When the value of the heat source is below the warm and cold fluids, the percentage will be negative.Although they are interesting inferences, even so it cannot be extended for a simple form of the synchronization of the Digital Twin counterparts.

LMTD remodeled

A remodeling of the "Model" mechanism was performed by changing the brackets and multiplying the expression for -1, thus changing the origin and the signal of exit of the calculation and obtaining values which better describe the behavior for a kinematic analysis of the Digital Twin synchronization. Some peaks of reflexion which are in the cutting line between 2 thousand per cent to 8 thousand per cent, both for more and for less were inserted in Fig. 2, the outcomes were generated from "Remodel", equation 4, depending on elapsed time.

$$Remodel = \left[\frac{(TCPU - Tin) - (TCPU - Tout)}{Ln\frac{(TCPU-Tin)}{(TCPU-Tout)}}\right] \times (-1) \tag{4}$$

At least four empirical regularities were found:

1. When the entering values Tin are higher than the exit values Tout the Remodel calculation outcome is negative.
2. When the entering values Tin are lower than the exit values Tout the Remodel calculation outcome is positive.
3. In 418 s time the highest percentage value was obtained which was 8.7808920%. In 418 s time the reference value TCPU was more distant than the Tin and Tout.
4. In the time interval of 540 to 600 it is observed that values Tin and Tout get close to the reference value TCPU the approach to zero occurs in chart of Fig. 2.Then at least three abductive inferences can be done.

1. If the values Tin, Tout and TCPU keep approximately or exactly equal one can say that it is almost synchronized or entirely synchronized.
2. In case one maintains a determined variation of temperature in a determined time based on the chart in Fig. 2 it would be enough to step up or step down Tin and Tout values according to a setpoint previously established.
3. The higher the percentage value of TCPU reference, the more distant it will be to Tin and Tout values.

The SyncLMKD method

The empirical regularities and inferences are relevant ideas for the mobile robotics domain specially for the synchronization of Digital Twin entities. The modeling which follows was inspired in Remodel method, as

- *Kvirt*: - Kinematics position of the virtual counterpart.
- *Kreal* - Kinematics position of the real counterpart.
- *Kideal* - Kinematics position of the idealized position (dynamic target).
- *SyncLMKD* - Log Mean Kinematics Difference of Synchronization.The log mean kinematics difference (LMKD) range can be written as

$$SyncLMKD = \frac{(Kideal - Kvirt) - (Kideal - Kreal)}{Ln\left|\frac{Kideal-Kvirt}{Kideal-Kreal}\right|} \tag{5}$$

Starting from the same assumption previously contextualized, the SyncLMKD method is suitable when the target position is known, independently to be static or dynamic. But it would not be simply to take a mean of kinematics movement differences because that position difference between virtual and real robot varies along the planned path. So, to use log mean kinematics difference of Synchronization (SyncLMKD method) is the most suitable one to calculate the position transfer of real and virtual robots.

Figure 3 presents two possibilities of the SyncLMKD method coupling, measurement of efficiency and the adjustment of synchronization (Effect), also some requirements of the DT were considered[23] , sensor fusion and artificial intelligence staying out of the approach which are not in bold.

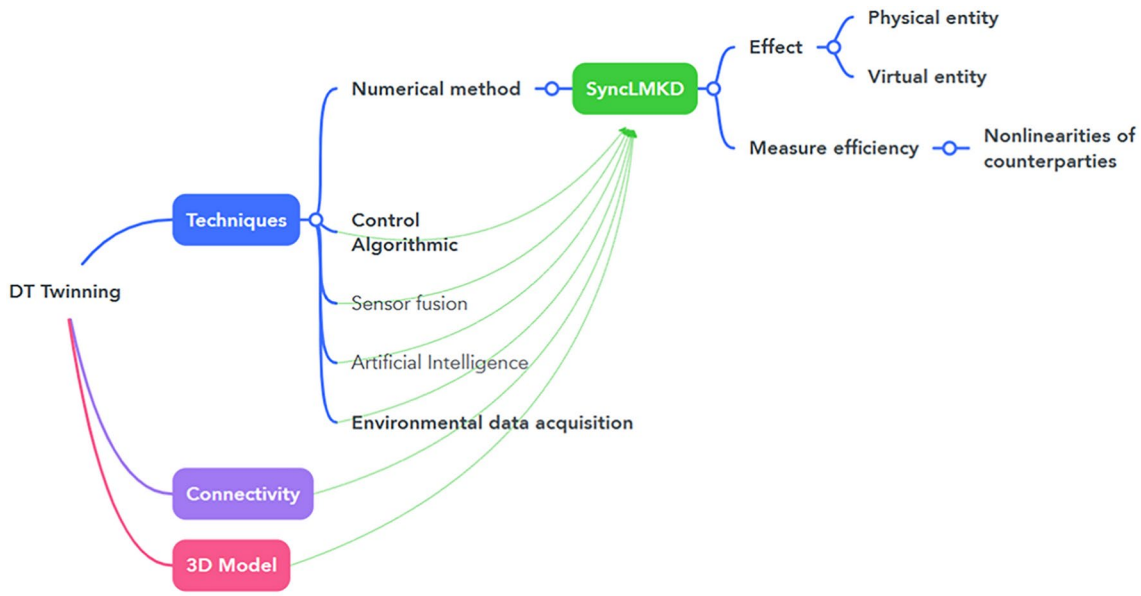


Fig. 3. Delimitation of SyncLMKD in the DT area.

The methodology of SyncLMKD Method is presented in Fig. 4 starts at receiving the Coord signal after it is checked which counterpart is ahead of the other. This is possible by the positive and the negative signals of the coordinates percentage no matter whether y axis or x axis. The adjustment of angular speed comes from the receiving signal of CoordX control and the linear speed adjustment comes from the receiving signal of CoordY control. The linear synchronization process and the angular one occur in three stages depending on the percentage received . Initially, stage 1, proportional effect - PE of 0.02 is applied when the distance between the counterparts are equivalent in meters. If the relation is about centimeters, the integrative effect is applied - IE of 0.001. And in case the approximate relationship is near millimeters or thousandths of millimeters the derivative effect will be applied - DE of 0.01. The addition of the weight term, which is a variable value and depends on the detection of pre-configured speeds and initial distances between the counterparts and the planning path,

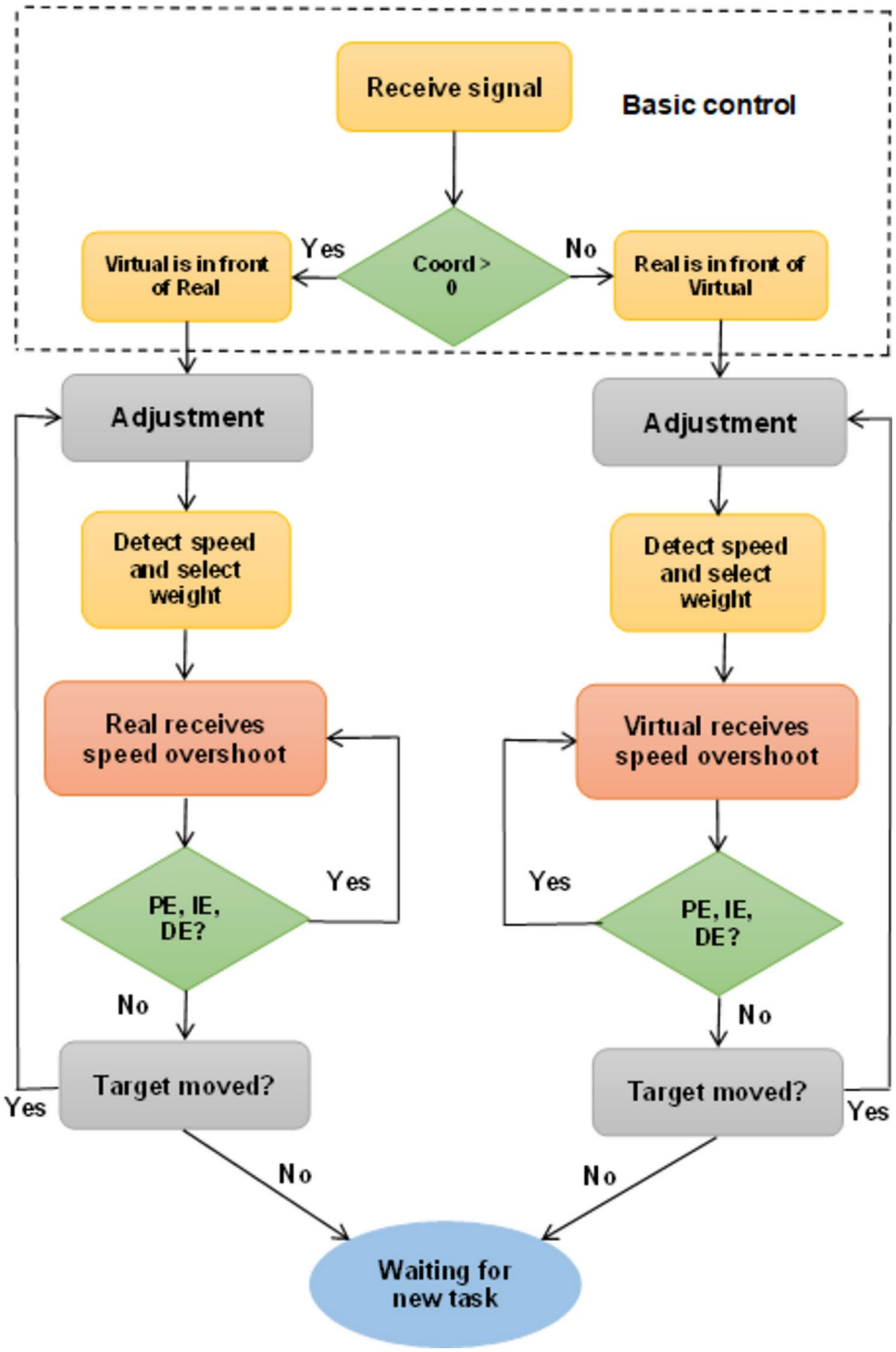


Fig. 4. Methodology of SyncLMKD method.

hampers directly in the output speed over-signals of the counterparts where greater weight is required when the distance is larger than the elements. In tests without the insert of weights, a delay was noticed in getting the intersection. With the insert of weight, there was a decrease of the time of intersection. The detected effects and weights are related as in equation (6).

$$\text{adjustment} = \left(\frac{1}{\text{coord}} + \text{weight}\right) \times \text{effect}. \tag{6}$$

After the synchronization has reached the counterparts and also the counterparts and the target of the dynamic planning, it turns out, for a certain moment, whether still there is the occurrence of planning movement, whether there is not, the current control flow is close and it waits for another task (aim) of synchronization.

Evaluation

Figure 5 presents details of the implementation of the CoppeliaSim simulation scene which is the representation of Digital Twin application. The synchronization technique and method control is externally applied through Robot Operating System (ROS). Taking into account that the calculations of SyncLMKD use only x and y coordinates, indicators 1, 2 and 3 of Fig. 5 show three separate surfaces of displacement which are aligned to the z axis in zero, centrally. So that Indicator 1 is the surface of the virtual robot, indicator 2 is the surface of the dynamic target and indicator 3 is the surface for the displacement of the representation of the real robot, by understanding that each element has a displacement surface equal to the other. For an overall view, displacement surfaces 2 and 3 were configured transparently. The counterparts, Robot1 (in red representing the virtual robot) and Robot2 (in blue representing the real robot), indicator 5. The dynamic target "Path_Planning", indicator 6, uses the example of scene "mobileRobotPathPlanning.ttt", of CoppeliaSim itself which was configured transparently too. The "lmcd_calculations", indicator 7 and 9, receive the percentage calculated from the SyncLMKD method for graph plotting. During the simulation running, position data concerning time were printed in "CoppeliaSim:info" exit, indicator 4.

Data used for evaluation came from[13] and[14] papers. In[13] the synchronization technique had not yet been implemented; only fixed speeds were established therefore the real counterpart (robot in blue) was behind the virtual counterpart (robot in red); 5 cm/s speed was prefixed for robot in blue and for the one in red 4 cm/s. As long as the robot in blue reached the one in red and there were the interaction and signal change, positive and negative, between the counterparts, the speed of the robot in blue was reduced for the same of the one in red, that is, 4 cm/s. The first relation among the exit percentage of SyncLMKD (CoordY, CoordX), the displacement between the counterparts and the dynamic target is presented in Fig. 6. The initial coordinates for the virtual counterpart (circle in red) are x=2.75; y = 2.57 and for real counterpart (circle in blue) are x = 1.86; y = 1.71, those coordinates are repeated for 6, 7 and 8 simulations, therefore the initial percentages are the same, 28% for

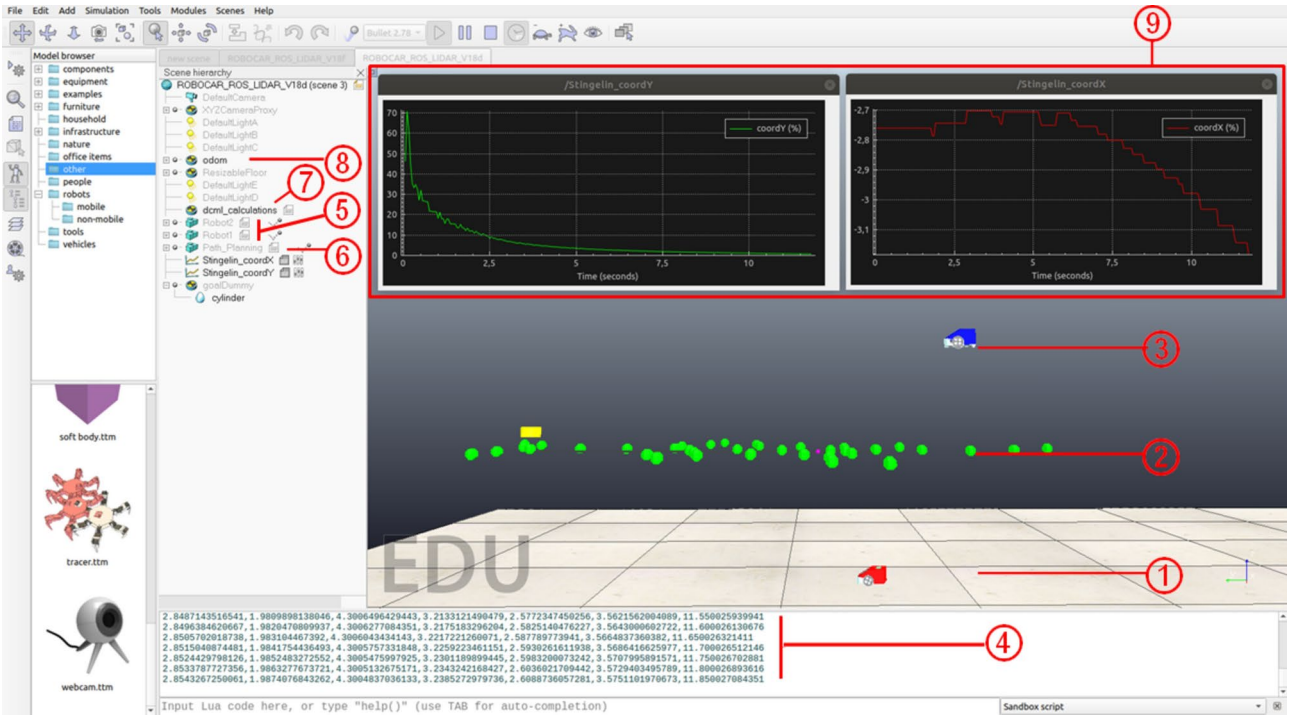


Fig. 5. CoppeliaSim simulation scenario.

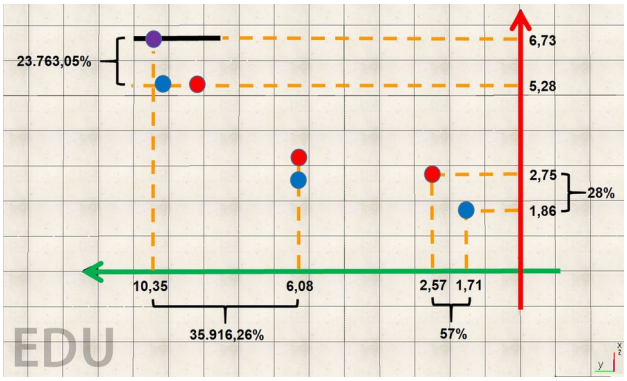


Fig. 6. Fixed speed in counterparts.

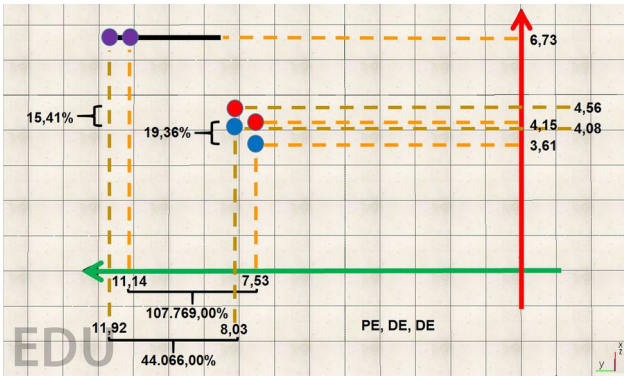


Fig. 7. Speed with (PE, DE, DE) adjustment at counterpart.

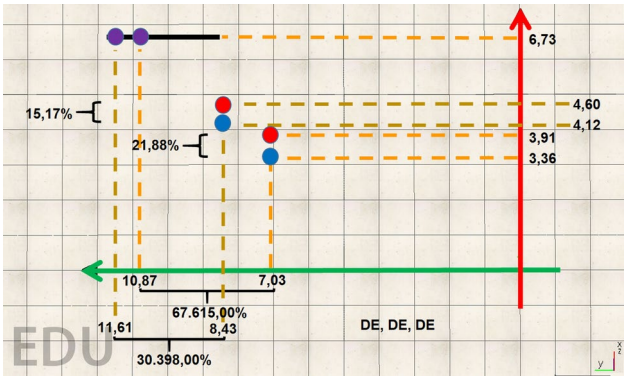


Fig. 8. Speed with (DE, DE, DE) adjustment at counterpart.

the exit of CoordX and 57% for the exit CoordY. The dynamic target of the planning path (circle in purple) has initial coordinates of x = 6.73; y = 8.55 and ends of x = 6.73; y = 11.92 and fixed speed of displacement of 2 cm/s. The intersection for CoordX occurred in 86.70 s time of position x = 5.28 axle. The moment of intersection for CoordY the dynamic target was at position of y = 10.35 axle and generated the percentage of 35.91626%. And in the moment of intersection for CoordX the dynamic target was at position x = 6.73 once its displacement is only linear without angular interference and generated the percentage of 23.76305%.

Figures 7, 8, 9 performed the same experiment of Fig. 6 but with the addition of the overload of adjustment of the speed synchronization technique[14]. The prefixed speed in the counterparts of those simulations was of 4 cm/s, added to the adjustment value of overload activation function according to assigned profile (PE, DE, DE), (DE, DE, DE) and (PE, IE, DE). Two interesting evaluations and which are correlated to the previsibility of SyncLMKD method are:

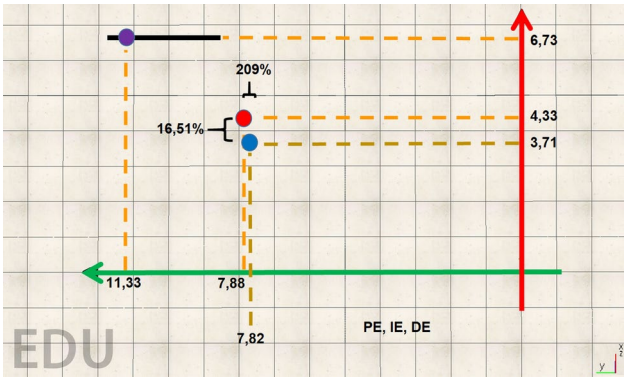


Fig. 9. Speed with (PE, IE, DE) adjustment at counterpart.

Virtual Y-axis	Real Y-axis	Planned Y-axis	SyncLMKD CoordY
5.3665	5.3668	11.5966	−158,964
5.3707	5.3705	11.5966	164,017
6.3665	6.3668	11.5966	−112,026
6.3707	6.3705	11.5966	115577
7.3665	7.3668	11.5966	−73,280
7.3707	7.3705	11.5966	75,593
8.3665	**8.3668**	**11.5966**	**−42,726**
8.3707	**8.3705**	**11.5966**	**44066**
9.3665	9.3668	11.5966	−20,364
9.3707	9.3705	11.5966	20,994
10.3665	10.3668	11.5966	−6194
10.3707	10.3705	11.5966	6379
11.3665	11.3668	11.5966	−216
11.3707	11.3705	11.5966	220

Table 1. Relation between the distance and the percentage.

Moment	Difference	Percentage
1	3.61	107.76900
2	3.89	44.06600

Table 2. Distancing between counterparties and target (PE, DE, DE).

1. the relation between the element distances and indication of percentages.
2. the approximation of the counterparts at simulations of Figs. 7 and 8 got a bigger time for the intersection when compared to simulation of Fig. 6 which performed an intersection time reduced; in the first case, gradual adjustment was observed, fractional, of synchronization.

About item 1, a comparison was generated with the percentage of intersection occurring at the second moment of the profile (PE, DE, DE). Table 1 presents the relation between the distance and the percentual value of exit for CoordY; the percentual values of the second moment of the profile (PE,DE,DE) were centrally inserted and are in bold in Table 1. The whole part of the counterparts position was incremented as well as was decremented. The value of axle position y for planning was kept the same for the calculation of SyncLMKD method at CoordY. It is observed that when closer the counterparts are of the planning, lower is the exit percentage and the more distant the counterparts are of the planning bigger is the exit percentage. Independent of the movement and speed the elements are, the perception of distance via percentages is constant and precise.

Comparing moment 1 and moment 2 among simulations (PE, DE, DE), (DE, DE, DE), Tables 2 and 3, it is possible to note that the difference of distance of 3,61 increased to 3,89 while the percentages decreased demonstrating a reverse relation to simulation (PE, DE, DE). In such simulation (DE, DE, DE) the relation is direct and both increase.

Moment	Difference	Percentage
1	3.84	67.615,00
2	3.08	30.398,00

Table 3. Distancing between counterparties and target (DE, DE, DE).

Moment	Difference	Percentage
1	0.0001	107.76900
2	0.0002	44.06600

Table 4. Distance between counterparties (PE, DE, DE).

Moment	Difference	Percentage
1	0.0002	67.61500
2	0.0002	30.39800

Table 5. Distance between counterparties (DE, DE, DE).

Moment 2	Time (s)	Millisec	Sampling range
PE, DE, DE	68.85 to 73.10	86	−505 to −42,726 and 44,066 to 512
DE, DE, DE	70.3 to 73.35	62	−508 to −30,398 and 19,117 to 505

Table 6. Short fractional propagation.

So we sought to understand the reason why the reverse relation, it refers specifically to the distance between the counterparts, as observed at Tables 4 and 5 when 107 thousand percentages were generated, the distance between the counterparts was of 1 ten thousandth, while in the other relations of distance was of 2 ten thousandth. Then there are two inferences of exit and the relation of distance, one is proportional between the counterparts and the dynamic target (global) and another between the counterparts themselves (local).

About item 2, the aspect of crumble the applied approximation speed via twin technique (twin technique is synonymous of synchronization in Digital Twin context) of SyncLMKD method which has been shown suitable taking into account that speed profiles when assumed integer absolute values is difficult to achieve, at the dynamic moment, to reach the mirroring without overtaking minimum distances as undertaken in the experiment which was around tenths of thousandths. Also, another note, it was at the second moment in which peaks occur for profiles (PE, DE, DE), (DE, DE, DE) occurred in a short moment and consequently a short fractional propagation for the adjustment and even so mirroring was reached. Table 6 presents data of short fractional propagation.

PE, IE, DE profile, Fig. 9, kept the mirroring with distributed kinematics difference which varied from centimeters up to millimeters after the initial approximation and the intersection of the counterparts did not occur.

Analysis and discussion

When analyzing the evolution of CoordY and CoordX in parallel, it becomes apparent whether the movement is being executed in a straight line. A straight-line movement relative to the planning path results in a stable percentage of CoordX, while a decrease in the percentage of CoordY indicates convergence towards the planned path.

In the case of curved movements along the planning path, although CoordY continues to converge, the percentage of CoordX ceases to stabilize and begins to increase. This increase indicates a divergence from the planning path relative to CoordX. Therefore, when a curve in the planning path is approaching at a close angle to the counterparts, it is necessary to initiate curve movement and adjust the angular speed of one wheel before CoordY reaches zero.

This evaluation is crucial, as the angular rotation characteristics of different mobile robots define the optimal percentages needed to perform the curve movement efficiently. Consequently, future research should focus on characterizing angular movements and determining the appropriate CoordY percentages required to initiate rotations during curved paths.

Two limitations were identified regarding the evaluation of movement and errors over time. First, for the SyncLMKD method to function correctly, the planned path must be ahead of or have a value greater than that of the counterparts; when one of the counterparts exceeds the planned path, whether in the x or y coordinates, the operation can result in a value of "not a number" (NaN). This effect occurs because a mathematical operation

Time (s)	Virtual (y-axis)	Real (y-axis)	Planned (y-axis)	CoordY (%)
106.75	10.989570	10.989572	11.927274	460.99846

Table 7. Timestamp with degraded percentage.

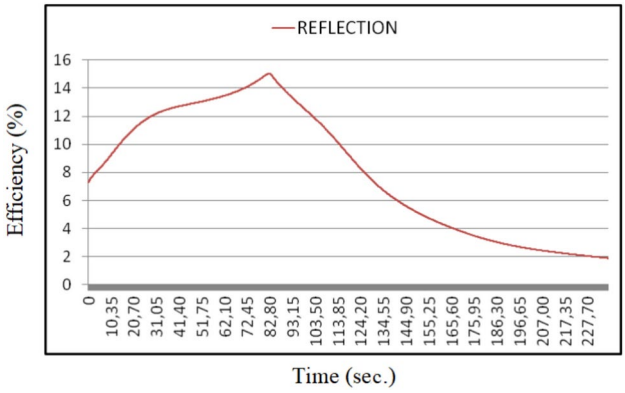


Fig. 10. Variation indicated by scattering and spread of points.

attempts to divide zero by zero, represented as Log(0/0)-a calculation undefined in mathematics, as indicated by the program compiler.

Robots and the planned path were intentionally positioned in the second quadrant of the scene, where all displacement surfaces use increasing and positive odometry values. This positioning was a practical solution for the experiment, as using negative values in the SyncLMKD calculations would also lead to NaN values.

An additional experiment was conducted to assess the degradation of the SyncLMKD method. This behavior aligns with expectations based on the LMTD formula, particularly when division occurs with "almost" absolute zero values. As a result, it was necessary to adjust Equation 6 and replace it with Equation 7. Instead of consistently using a numerator of 1, the numerators were modified to use values of 100, 10, and 1 depending on the proximity of the counterparts to the planned path. Specifically, a numerator of 100 is used when the counterparts are farther from the planned path, while a numerator of 1 is applied when they are closer.

This adjustment allowed for the acquisition of valid values during the experiment and facilitated the determination of minimum speeds to be added to the pre-set speeds of the counterparts when they are in proximity to the planned path. This refined approach enhances the reliability of the SyncLMKD method, improving its performance under varying conditions.

$$adjustment = \frac{1}{coord} \tag{7}$$

The analysis of that experiment with minimum and over signal speed and also of others conducted show that the secure distance between the counterparts must be up to decimal of ten thousandths because below that the term "coord" can bring values which are undetermined, i. e., the degradation. Table 7 presents CoordY degraded value at millionths (difference in distance between the counterparts of 0.000002) in time of 106.75 s of Slingshot Effect - SE simulation.

Therefore by establishing the ten-thousandth tolerance it prevents indexed values and the adjustment of the synchronization speed and consequently movements not determined. The datasets of all simulations, scenes (representation) and the source codes used (controllers), also the step to step for the deployment of packages are available in "Code and Data availability" an open share link.

Performance metrics

Grieves, Digital Twin concept creator, proposed in 2017 a new methodology based on the adaptation of Turing method, known as "The imitation game",[24]. That adaptation aimed to evaluate and measure via human observator the degree, or level, of synchronization of Digital Twin application. So, this section presents an unprecedented graphic metric which can be used to evaluate and measure the efficiency of the synchronization for Digital Twin applications at the human observator place.

Graphically the fractional logarithmic propagation, exits of CoordY, CoordX of SyncLMKD method demonstrate the variation, synchronization and stabilization of digital twins along the evaluated time.

Classical Proportional, Integral and Derivative control - PID was entrained in both virtual counterparts in order to observe its efficiency via SyncLMKD method,[13]. Figure 10 refers to displacement in y with PID control. The initial coordinates to the simulation with PID were different to the previous ones, as the virtual counterpart

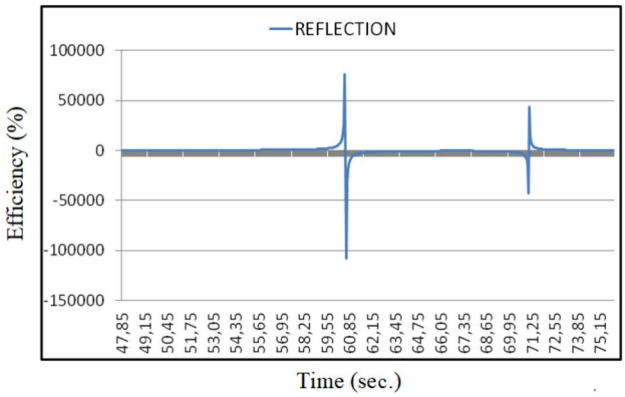


Fig. 11. Synchronization indicated by the peaks.

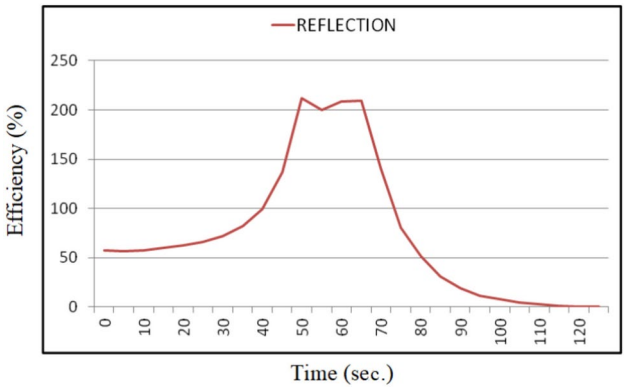


Fig. 12. Stabilization indicated by the horizontal part of the line.

(circle in red) to x = 3.94; y = 1.48 and to the real counterpart (circle in blue) are x = 3.06; y = 0.62 and of the Path_Planning x = 4.25; y = 3.07. The graph line points are scattered and spread, from 8 to 15%, that is, a kinematics variation occurred at that time of execution performance. It demonstrates that the synchronized displacement at y axle had low efficiency via PID control. At 82.80 s time and efficiency of 15% a crude descent starts. That occurs because Path_Planning reaches the destination position, remaining still allowing the PID control to recalculate its coordinates. In PID control simulation the SyncLMKD method coupling of speed adjustment did not occur; it occurred only the measurement of PID controller efficiency after that the dynamic planning path.

Figure 11, specifically at simulation (PE, DE, DE), the time of simulation was limited, encompassing from 47.75 to 75.75 s, without imposing any restriction of peak interleaving. During that interleaving, it is evident that a saturation occurred, starting at the peak of the first moment and stretching to the second one causing a reduction in its amplitude. That notes shed light on SyncLMKD method previsibility. If the intersection between the counterparts is generated, it means that the synchronization occurred and valuable information about the proximity of the planning path can be obtained from exits. For each peak generated there is a correlation of percentage and relative distance specific between the counterparts and the planning path (dynamic target).

Figure 12 presents PE, IE, DE profile which allows an approximation without the intersection but with percentage stabilization. Initially the counterparts were out of synchronization at zero seconds time, they got closer from 0 to 50 s time, kept a minimum distance from centimeters to millimeters of variation by 50 to 70 s time e at last with the approximation of the dynamic target jointly remained synchronized from 70 to 120 s period of time.

Generally the sum of errors of Euclidean distances are used to apply for gains; in the present case the exit CoordY value was used for effects (PE, IE, DE), (PE, DE, DE) and (DE, DE, DE) in equation (6).

Conclusion

Over the past decade, the exponential growth of the Digital Twin (DT) concept underscores the need for further investigation into its specific properties. A noticeable gap exists in the research focused on the counterparts of mobile robots, with only a limited number of studies dedicated to the accurate formulation of mathematical models. Most existing approaches typically rely on artificial intelligence strategies to subtract position differences, which, while valid, do not fully address the fundamental requirements for DT applications.

Their proximity during displacement characterizes Fidelity or the reflection of DT counterparts. Variations measured in meters are easily perceptible, while changes at more minor scales, such as centimeters and millimeters, lead to reduced perception of reflection and result in more rigid synchronization. In this context, it has been observed that the speed required for synchronization is inversely proportional to the value of the Coord 6 term in the equation. Additionally, incorporating a weight can facilitate a reduction in the synchronization response time, and decomposing the approaching states into eight tracks enhances the effectiveness of synchronization at optimal moments.

In the experiments conducted, it was determined that the maximum percentage could reach up to 200,000. However, subsequent simulations achieved a peak of 107,000-increased distances between elements correlated with higher percentage values. When the simulation distances were multiplied by a factor of ten, expanding them significantly, the percentage surged to the million mark. This finding highlights the need for more in-depth studies on the relationships between distances among elements and their impacts on synchronization percentages across various layouts.

Data availability

The datasets generated and/or analysed during the current study are available in the [2024] repository, [https://github.com/facardoso-sudo/2024].

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References

1. Chen, T., Peng, L., Yang, J. & Cong, G. Analysis of user needs on downloading behavior of English vocabulary apps based on data mining for online comments. *Mathematics* **9**, 1341. https://doi.org/10.3390/math9121341 (2021).
2. Chen, T. et al. Monitoring and recognizing enterprise public opinion from high-risk users based on user portrait and random forest algorithm. *Axioms* **10**, 106. https://doi.org/10.3390/axioms10020106 (2021).
3. Shahgholian, S., Akhavan, M., Kamrani, V. & Ganjefar, S. Design an omnidirectional autonomous mobile robot based on non-linear optimal control to track a specified path. *IET Control Theory Appl.* (2024).
4. Yu, M. et al. Hand medical monitoring system based on machine learning and optimal EMG feature set. *Pers. Ubiquit. Comput.*[SPACE]https://doi.org/10.1007/s00779-019-01285-2 (2019).
5. Luo, B., Sun, Y., Li, G., Chen, D. & Ju, Z. Decomposition algorithm for depth image of human health posture based on brain health. *Neural Comput. Appl.* **32**, 1–16. https://doi.org/10.1007/s00521-019-04141-9 (2020).
6. Tiseo, C., Merkt, W., Wolfslag, W., Vijayakumar, S. & Mistry, M. Safe and compliant control of redundant robots using superimposition of passive task-space control-lers. *Nonlinear Dyn.* **112**, 1023–1038 (2024).
7. Zager, M., Sieber, C. & Fay, A. Towards semantic interoperability: An information model for autonomous mobile robots. *J. Intell. Robot. Syst.* **110**, 1–19 (2024).
8. Li, X. et al. Dynamic path planning of mobile robots using adaptive dynamic programming. *Expert Syst. Appl.* **235**, 121112 (2024).
9. Deolia, V. K. et al. Comparative performance analysis of robust and adaptive controller for three-link robotic manipula-tor system. *J. Electr. Eng.* **75**, 275–284 (2024).
10. Hassan, I. A., Abed, I. A. & Al-Hussaibi, W. A. Path planning and trajectory tracking control for two-wheel mobile robot. *J. Robot. Control* **5**, 1–15 (2024).
11. Pauca, G.-S. & Caruntu, C.-F. Mpc-based dynamic velocity adaptation in nonlinear vehicle systems: A real-world case study. *Electronics* **13**, 2913 (2024).
12. Stingelin, F., Renaux, D. P. B. & Lima, C. R. E. Uso da plataforma arduino e do software plx-daq para o ensino do resfriamento líquido em sistemas eletrônicos. *Educação Profissional e Tecnológica em Revista* **6**, 32–43 (2022).
13. Cardoso, F. S., Cantieri, A. R. & de Oliveira, A. S. Controle cinemático de sincronização para as contrapartes do gêmeo digital através do novo método synclmkd. In *Simpósio Brasileiro de Automação Inteligente-SBAI* (2023).
14. Cardoso, F. S., Rohrich, R. F. & de Oliveira, A. S. The twinning technique of the synclmkd method. In *Robotics, Computer Vision and Intelligent Systems*, 426–440 (eds Filipe, J. & Röning, J.) (Springer Nature Switzerland, 2024).
15. Kuts, V., Cherezova, N., Sarkans, M. & Otto, T. Digital twin: Industrial robot kinematic model integration to the virtual reality environment. *J. Mach. Eng.* **20**, 53–64 (2020).
16. Liang, C.-J., McGee, W., Menassa, C. & Kamat, V. Bi-directional communication bridge for state synchronization between digital twin simulations and physical construction robots. In *Proceedings of the International Symposium on Automation and Robotics in Construction (IAARC)* (2020).
17. Müller, M. S., Jazdi, N. & Weyrich, M. Self-improving models for the intelligent digital twin: Towards closing the reality-to-simulation gap. *Ifac-Papersonline* **55**, 126–131 (2022).
18. Liu, X. et al. Genetic algorithm-based trajectory optimization for digital twin robots. *Front. Bioengineering Biotechnology* **9**, 793782 (2022).
19. Bergman, T. L., Lavine, A. S. & Queiroz, E. M. *Fundamentos de Transferência de Calor E de Massa* . (Grupo Gen-LTC, 2000).
20. Kay, J. M. & Nedderman, R. M. *Fluid mechanics and transfer processes* (CUP Archive, 1985).
21. Wertheimer, M. *Max Wertheimer Productive Thinking* (Birkhäuser; 1st ed. (2020).
22. Woodward, J. Making things happen: A theory of causal explanation. *Making Things Happen: A Theory of Causal Explanation*[SPACE]https://doi.org/10.1093/0195155270.001.0001 (2003).
23. Rasheed, A., San, O. & Kvamsdal, T. Digital twin: Values, challenges and enablers from a modeling perspective. *IEEE Access* **8**, 21980–22012 (2020).
24. Grieves, M. & Vickers, J. *Digital Twin: Mitigating Unpredictable, Undesirable Emergent Behavior in Complex Systems, 85–113* (Springer International Publishing, 2017).

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Author contributions

All authors contributed to the conception, writing and editing of this article.

Declarations

Competing interests
The authors declare no competing interests.

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